

Embodied Spatial Simulation: Enabling Embodied Interaction with Human Motion Simulation in Extended Reality

Yanni Mei, Jonas Wombacher, Wen-Jie Tseng, Veronika Krauß, and Jan Gugenheimer



Fig. 1: A) The user, wearing an AR headset, was planning fire evacuation in situ. (B) She created virtual agents representing the occupants. (C) Virtual fires were ignited. (D) The user embodied the smallest agent to assess the reachability of the fire alarm for everyone. (E) After pressing the fire alarm, the agents attempted to escape but headed toward the incorrect exit. (F) Taking the visual perspectives of one agent, the user found that the agent was colorblind and unable to distinguish the green emergency sign. (G) The user added a more visible sign virtually. (H) The simulation was rerun to evaluate the solution, the agents ran toward the correct exit.

Abstract—Human Motion Simulation (HMS) predicts and visualizes how virtual agents move and behave in various scenarios, aiding decision-making. However, it does not communicate the experience behind agents' behavior. To address this, we present Embodied Spatial Simulation (ESS), a concept combining augmented reality (AR) and HMS to support in-situ decision-making. It enables an XR user to create “what-if” scenarios, simulate virtual agents' behavior in situ, and experience the scenarios from their perspectives through embodiment. To demonstrate ESS, we developed a proof-of-concept prototype featuring two scenarios (e.g., fire evacuation planning). From expert walkthroughs (N=4) and a survey (N=38) with different professions, we found that incorporating embodiment into simulations can change users' intentions to use simulation outcomes. Rather than directly obtaining solutions from the results, users showed more interest in understanding the experience behind simulated behavior and using this insight to shape decisions. We discuss the potential of ESS to support future human-centered design practices.

Index Terms—Extended Reality, Human Motion Simulation

1 INTRODUCTION

- Yanni Mei is with TU Darmstadt, Darmstadt, Germany.
E-mail: yanni.mei@tu-darmstadt.de
- Jonas Wombacher is with TU Darmstadt, Darmstadt, Germany.
E-mail: jonas.wombacher@tu-darmstadt.de
- Wen-Jie Tseng is with Sorbonne Université, Paris, France.
E-mail: wen-jie.tseng@sorbonne-universite.fr
- Veronika Krauß is with Hochschule Ansbach, Ansbach, Germany.
E-mail: veronika.krauss@hs-ansbach.de
- Jan Gugenheimer is with TU Darmstadt, Darmstadt, Germany.
E-mail: jan.gugenheimer@tu-darmstadt.de

Manuscript received xx xxx. 201x; accepted xx xxx. 201x. Date of Publication xx xxx. 201x; date of current version xx xxx. 201x. For information on

Human Motion Simulation (HMS) forecasts how future “what-if” scenarios influence the motion behavior of virtual agents [1, 11], such as walking routes in public spaces [18, 64] or postures adapting to furniture [39]. HMS can generate human motion data to predict human reactions when developing human-machine/environment systems, and hence support decision-making [1]. However, using HMS to inform decisions still remains challenging. Human motion behavior is strongly shaped by surrounding physical or social contexts. Without these contextual factors, interpreting HMS data becomes difficult [48, 62]. While primarily focusing on supporting users with simulated data, such as showing the most time-efficient walking route, existing HMS tools

obtaining reprints of this article, please send e-mail to: reprints@ieee.org.
Digital Object Identifier: [xx.xxxx/TVCG.201x.xxxxxx](https://doi.org/10.1109/TVCG.2021.3088888)

fail to communicate how an individual experiences the walk, such as environmental perceptions, interactions, or comfort [3, 41]. These experiential factors can be crucial in decision-making across fields like design, architecture, and social robotics, which now prioritize human-centered considerations [41, 56, 68]. Thus, we seek a new approach to understanding and experiencing HMS and adding human-centered value to decision-making.

With recent advances in Head-Mounted Displays (HMDs), Extended Reality (XR) is becoming a promising tool for visualizing, interacting, and experiencing human motion behavior data. By enabling situated visualization, XR can display human motion data in the physical environment where it was captured, facilitating direct understanding and spatially aware analysis [10, 48, 62]. Additionally, XR can augment people's embodied experience to understand their behavior. For instance, through an XR HMD, people can directly experience being a small person and interacting with surroundings from a lower eye level [56].

Using XR's capabilities in situated visualization and experience augmentation, we propose the concept of **Embodied Spatial Simulation (ESS)**, enabling an XR user to **run HMS in situ** and then **experience the simulation results** to inform decisions. ESS provides an immersive environment for an XR user to configure "what-if" scenarios in situ, simulate how virtual agents react to future scenarios in real-time, and then embody the agents to experience the world through their eyes and bodies. To support ESS, we propose three components of the concept: (1) the direct manipulation of the XR environment to query the simulation model (*Embodied Interaction*), (2) the immediate generation and display of simulation results through the behavior of virtual agents within the physical space (*In-situ Simulation*), and (3) the ability for the user to embody virtual agents, experience the scenario through their eyes and with their body capabilities (*Agent Embodiment*).

We illustrate these three components through a scenario (Fig. 1) showing how ESS is used in fire evacuation planning. A security staff using XR is situated within the room to plan fire evacuation (Fig. 1A). To build the emergency scenario, she created virtual agents representing future occupants (Fig. 1B) and simulated a potential fire situation (Fig. 1C). To ensure the fire alarm could be pressed before the agents escaped, she had to assess its reachability. She controlled the body of the smallest virtual agent, attempting to press the fire alarm (*Embodied Interaction*) (Fig. 1D). With the fire alarm pressed, the virtual agents attempted an escape route but used the wrong exit (*In-situ Simulation*) (Fig. 1E). To investigate, she took the visual perspective of a virtual agent, revealing that the agent could not perceive the green emergency sign due to colorblindness (*Agent Embodiment*) (Fig. 1F). She then adopted a more visible sign and reran the simulation to assess the new solution (Fig. 1G).

To demonstrate this concept, we implemented an XR prototype presenting two scenarios to support decision-making, such as planning fire evacuation and organizing VR play sessions for children. We conducted scenario walkthroughs with four XR researchers. Our findings indicated that *Agent Embodiment* helped participants identify problems from multiple perspectives and focus on the reasoning behind the simulation outcomes. We then performed an online survey (N=38) to assess the concept with individuals from various occupations. Participants were asked to create scenarios for both *In-situ Simulation* and *In-situ Simulation with Agent Embodiment* that could be incorporated into their daily work. We found that, using only *In-situ Simulation*, participants mainly sought actionable decisions directly from simulation results. By adding *Agent Embodiment*, we observed a shift in participants' focus toward reflecting on the experiences behind the simulated behavior, seeking insights to shape decisions. We also discovered diverse scenarios where HMS was not applied traditionally.

This paper demonstrates three main contributions:

- We introduced the concept of *Embodied Spatial Simulation (ESS)*, a combination of XR and HMS to support in-situ decision-making. It consists of three components: *Embodied Interaction* to query the simulation model, *In-Situ Simulation* to immediately run and see simulation in-situ, and *Agent Embodiment* to experience simulation results.
- We present a proof-of-concept implementation enabling XR users to configure "what-if" scenarios, simulate virtual agents' navigation behaviors, and embody the agents to experience their perspectives and physical capabilities. We applied this prototype to two scenarios.
- We report the results from pilot scenario walkthroughs (N=4) and an online survey (N=38) to evaluate the ESS concept. We discuss how ESS can help users reflect on the underlying experiences behind simulated behaviors, rather than simply relying on them to make decisions.

2 RELATED WORK

This section presents related work about Human Motion Simulation for decision-making, Human Motion Data in XR, and Transformed Embodiment Experience.

2.1 Human Motion Simulation for Decision-Making

Human Motion Simulation (HMS) uses mathematical models to predict how virtual agents with defined profiles (e.g., age) respond to future scenarios, supporting the design of human-machine and human-environment systems [1, 11, 39, 40, 47]. Integrated into commercial software, HMS allows users to configure agent profiles and "what-if" scenarios (e.g., 3D architectural models) to query the models for idea testing [40]. Simulated behaviors are visualized through animated avatars and data representations (e.g., heatmap) for analysis [1, 40, 64]. However, these outputs are typically viewed on desktops, which detach analysts from the context-rich environments that critically influence motion behavior [10, 48].

While HMS tools usually optimize for efficiency (e.g., suggesting the shortest walking path) [1, 18, 21], these techniques often overlook the experiential aspects behind simulated behaviors (e.g., social comfort during walking [2, 41]). These human-centered factors are increasingly valued in domains like architecture and product design [4, 41, 55, 68]. For example, Lee et al. showed that walking paths influence privacy comfort and social connectedness. They proposed the concept of "spatial-socio comfort" for designing human-building interactions, in contrast to most tools focusing only on path optimization [41]. This highlights the need to balance efficiency-focused simulations with human experience, which is often tacit and best understood through direct experience [2, 3, 41].

With ESS, we aim to allow people to directly view and interact with HMS in the relevant physical context using XR. We also aim to engage users in understanding the experience behind simulated behavior and bringing human-centered value into decision-making.

2.2 Human Motion Data in Immersive Visualisation

Capturing and visualizing captured human motion data can support analyzing human behavior and user experience [27, 48, 74]. Recently, AR is becoming a promising medium for visualizing and interacting with such spatial-temporal data [12]. Using situated visualization, AR integrates data representations seamlessly into the original physical environment where the data was collected [10, 48, 62], hence to support context-aware data understanding and analysis [16, 17, 35]. Additionally, motion data is usually embodied in humanoid avatars in AR. *AvatarAR* visualized virtual humans with walking path and eye gaze in a shop, allowing in-site analysis of consumer behavior [62].

In addition, AR also allows embodied interaction for manipulating human motion data [17]. Using motion tracking, AR enables users to directly manipulate data representations with natural bodily actions for data analysis (e.g., filtering) [12, 53]. For instance, Wagner et al. created a 3D virtual space-time cube that responds to hand gestures, allowing users to rotate and scale it for data querying [75]. Prior research demonstrated that embodied interaction with data representation can enhance insight exploration [26].

Similar with prior work, with ESS we aim to leverage AR's capabilities in situated visualization and embodied interaction to enhance the analysis of and interaction with HMS. Additionally, we go further by emphasizing AR's capability in embodiment, allowing users to directly experience the behavior data. We will elaborate in the next subsection.

2.3 Transformed Embodiment Experience in XR

Embodiment experiences represent how individuals perceive and interact with the world through their bodies [14, 15]. Thus, transforming one's embodiment experience to another enables one to empathize with another person, and even change one's perception [47, 72], emotion [69] and behavior [19, 76]. This concept is leveraged to make decisions for special populations such as kids [56], in the domains like design [56, 58], and education [5].

In the XR domain, transforming embodied experience mainly focuses on visual perspective and body ownership. Visual perceptions are transformed by capturing the first-person-view (FPV) of one person with wearable cameras and then streaming it through HMD [30, 31, 56]. Prior research also explores simulating vision limitations (e.g., cataract) on HMD for understanding individuals with eye diseases [37, 44, 77]. Through taking another's visual perspectives, viewers could better understand the sharer's spatial and social context [30]. Another XR approach to transform the embodied experience is enabling an XR user to inhabit in another virtual body. Inhabiting in another body representation can generate the illusion of being another individual [23] with different body sizes [8, 45, 55], genders [20] or skin colors [49, 60]. Users can also control the body to proactively interact with the environment and get direct proprioceptive feedback from the surroundings [32, 56].

XR enables direct engagement with others' embodied experiences. Recent HMS technologies simulate agent behaviors across various profiles [39, 40]. We aim to bridge these approaches by letting users embody virtual agents, understand the experiences of individuals represented by the virtual agents, and hence, support human-centered decision-making.

3 EMBODIED SPATIAL SIMULATION: CONCEPT AND PROTOTYPE

We introduce the concept of **Embodied Spatial Simulation (ESS)**, allowing users to simulate human motion with HMS and in XR for in-situ decision-making. Fig.2 illustrates a story to elaborate the concept: a curator altered the exhibition space, simulated the behavior of an audience in XR, and then embodied him to understand his viewing experience. We highlight three components of ESS to support decision-making:

Embodied Interaction to Configure Scenarios. Embodied Interaction lets users directly manipulate the environment or virtual agents to set up "what-if" scenarios, configuring set-ups such as agents' profiles, environment layouts, and events (Fig.2A). We encourage users to explore decisions by trying different scenarios.

In-situ Simulation to Explore Solutions. With a given "what-if" scenario, In-situ Simulation allows users to simulate the virtual agents' motion behavior in response to "what-if" scenarios, and immediately view the results through animated humanoid avatars (Fig.2B) in-situ. This aims to contextualize HMS data with physical environment to support data understanding.

Agent Embodiment to Evaluate Decisions. Given that the primary representation of HMS results revolves around virtual agents, ESS provides an Agent Embodiment feature. It enables users to embody simulated virtual agents, by taking their FPV or controlling their body to interact with the environment (Fig.2C,D). The user can experience the scenario from the agents' perspective and with their body capabilities. This functionality aims to support users in understanding the experience behind the simulated behavior.

Next, to demonstrate our ESS concept, we implemented an XR prototype, allowing an XR user to simulate the virtual agent's navigation behavior under various "what-if" (e.g., fire evacuation planning), and then embody these agents by taking their perspective or controlling their bodies. We first present the XR technical architecture (Sec.3.1), then detail the three interaction steps and their implementation (Secs.3.2–3.4).

3.1 XR Architecture

We implemented our prototype using Meta Quest Pro and Unity3D¹. We developed the video-passthrough AR with Meta SDK. To enable functionalities (e.g., perspective change in AR) that cannot be realized in video-passthrough, we overlaid a digital 3D reconstruction of the environment to the physical space in XR [29, 46]. To create it, we photo-scanned the environment using Polycam². The 3D reconstruction was automatically turned on when calling functions that cannot be realized in video-passthrough AR [46]: (1) *Taking the perspective of the agent*. Users can align their viewing location and orientation to the ones from the virtual agent without physical movement. This is possible because the 3D reconstruction has a comprehensive visual coverage of the environment [46]. (2) *Manipulating the environment using 3D construction*. User can resize the 3D construction to get a world-in-miniature (WIM) [71] in AR to analyze the environment with an overview (Fig.3). By changing the texture, the user can simulate vision with colorblindness (Fig.1F).

3.2 Step 1: Embodied interaction to Configure Scenarios

To query HMS models, Embodied Interaction allows users to set up "what-if" scenarios, by directly manipulating the environment with bodily actions. We provided users with two capabilities:

Create Virtual Agents. We allow the user to create virtual agents, by specifying the expected body size, location, and orientation. Users can point to the target location with the controller, drag upward to set the agent's scale, and then adjust its rotation using the controller (Fig.4).

Create Virtual Objects. We allowed the user to create and manipulate virtual objects required in future scenarios. Users could select an object on a virtual menu, and then place it by pointing to a location with the controller. Users could also manipulate them using natural bodily actions, such as grabbing the virtual fire alarm to reposition it (Fig.1D). The current implementation included various furniture and fire (Fig. 1C) that can be added to the scene. This prototype can be expanded in the future by incorporating a 3D asset database to meet the specific needs of scenarios.

3.3 Step 2: In-situ Simulation to Explore Solutions

With a given "what-if" scenario as input, we introduce *In-situ simulation* enabling users to run the simulation real time in-situ, as a way to quickly explore their solutions. We provide two capabilities:

Real-time In-situ Simulation After setting up the agents and scenario (e.g., four agents in a room on fire), users can start the simulation by clicking Oculus A button. Subsequently, each virtual agent would start navigation behavior responding to the scenario, such as running toward the exit during a fire. Furthermore, the relevant motion data (e.g., running route) would be displayed around the agents (Fig.5). All simulation outcomes were visualized in real-time in situ. In this prototype, we mainly focused on implementing navigation behavior simulation with NavMesh Path-Finding Plugin. We enabled the virtual agent to understand the scene by labeling the fire and furniture as "obstacles" to dynamically adjust its path to avoid collision, and labeling the exit as the "destination" to navigate toward. The walking, running, and idling postures were preset animations triggered by different navigation conditions of the agent. The agent's running speed was preset in NavMesh. This prototype used NavMesh as a lightweight simulation to support the core functions: letting users change "what-if" scenarios (e.g., fire location) and immediately see consequent results (e.g., navigation routes). These are key interactions in HMS tools. NavMesh is efficient for concept demonstration as our research goal. Additionally, to implement in-situ visualization, we used Spatial Anchor from the Meta SDK.

Analyze Simulation Outcome. Users could interact with data representations of the simulation outcome tailored to their analytical goals. Our prototype visualizes simulated motion history and allows users to scrub through trajectories via a timeline (Fig. 5). To provide

¹<https://unity.com/>

²<https://polycam.com/>

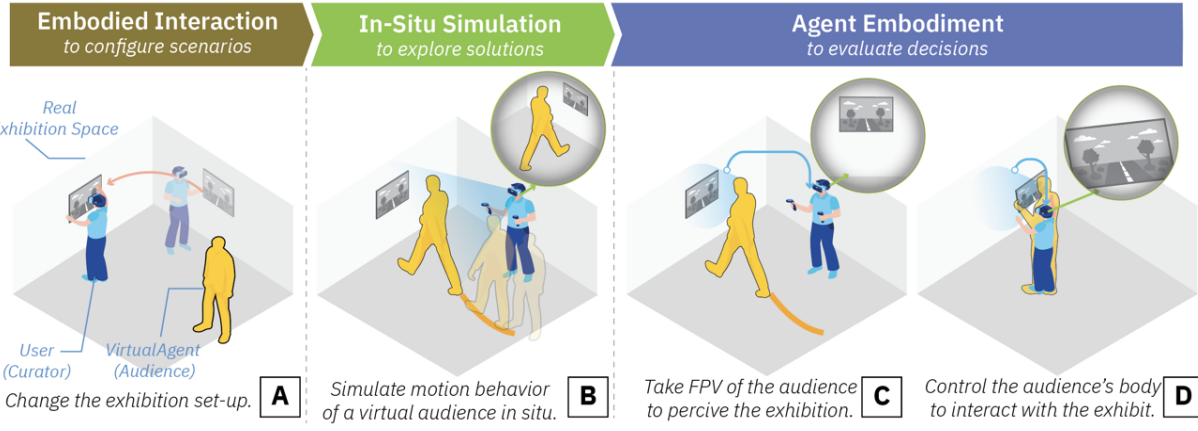


Fig. 2: ESS concept encompasses three components: (A) Embodied Interaction (B) In-situ simulation and (C&D) Agent Embodiment

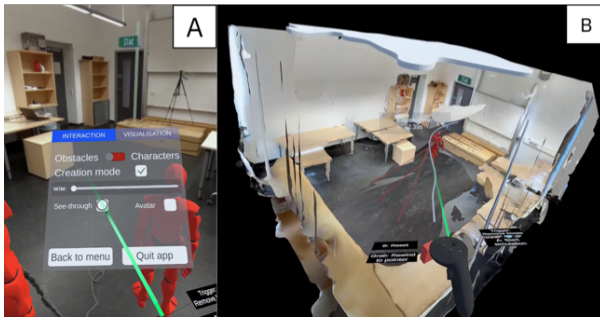


Fig. 3: (A) The user was situated in the video-seethrough AR mode, and then (B) turned on the WIM mode with the 3D reconstruction activated and video-seethrough turned off.

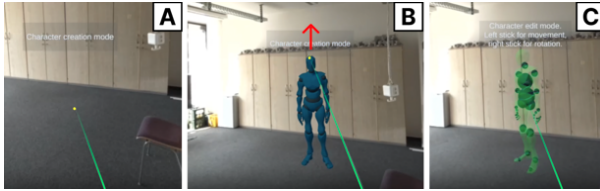


Fig. 4: To create an agent, the user (A) points to a location with the controller, (B) raise it to adjust scale, and (C) rotate for fine-tuning

an overview of the environment, we implemented a WIM (Fig.3) to analyze the whole environment. Users can activate the WIM through a graphical interface and resize it with a slider (Fig.3A).

3.4 Step 3: Agent Embodiment to Evaluate Decisions

With simulated agents around, *Agent Embodiment* enables users to embody these agents and experience the scenarios from their visual perspective and with their bodily capabilities. Our goal was to help users understand the experience behind the simulated behavior, such as spatial perceptions or comfort. We provided two functions:

Take Visual Perspectives. The user could take the FPV of the virtual agent. To trigger this, the user could point the controller at the target agent and then press the Oculus X button. While remaining stationary in the physical environment, the user's viewpoint seamlessly transitioned to match the position and orientation of the agent's eyes (Fig.1F). Without the necessity for physical movement, the user could concentrate on experiencing and interpreting the agent's viewpoint. To implement perception change in XR, as previously discussed in Section 3.1, the 3D reconstruction was automatically switched on when the

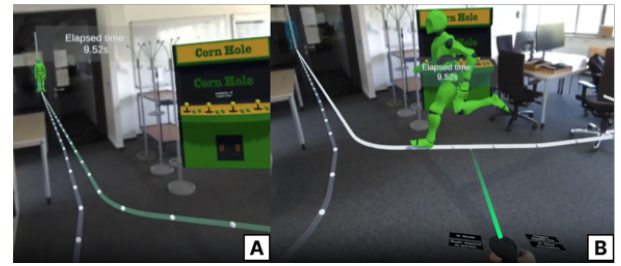


Fig. 5: (A) The trajectory indicated the navigation path of the virtual agent. (B) The user slid through the trajectory to check motion history

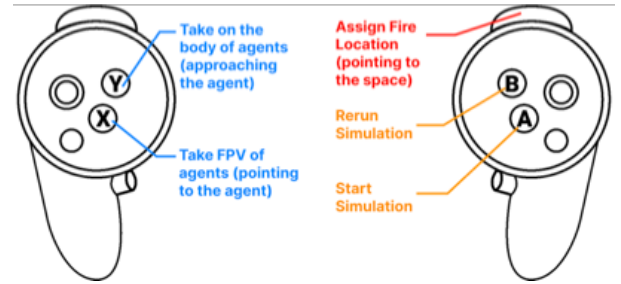


Fig. 6: The mapping between functions and button interactions.

user triggered the function. The position and orientation of the users' XR camera were adjusted within Unity3D, aligning it with the eyes of the agents.

Take Control Of Agent's Body. The user can also take full control of the virtual agent's body. Here, the user's body was transformed into the agent's representation in a different body size, so that the user was able to experience the body capability like limited arm reach or eye height (Fig.1D). To trigger this, the user needed to physically approach the same location as the agent and press the Oculus Y button. Subsequently, the agent's movements and posture would mirror those of the user, the user could interact with the environment through the virtual agent's body. To implement body control, we used *Animation Rigging* to map the users' posture to the agent's avatar. The users' posture was captured by the two XR controllers and headset.

All the functionalities are triggered as presented in Fig.6.

4 SCENARIO DEMONSTRATION

Using this prototype, we implemented two scenarios to demonstrate the versatility of ESS: fire evacuation planning and VR play session preparation. We will first illustrate each scenario. Next, we explain

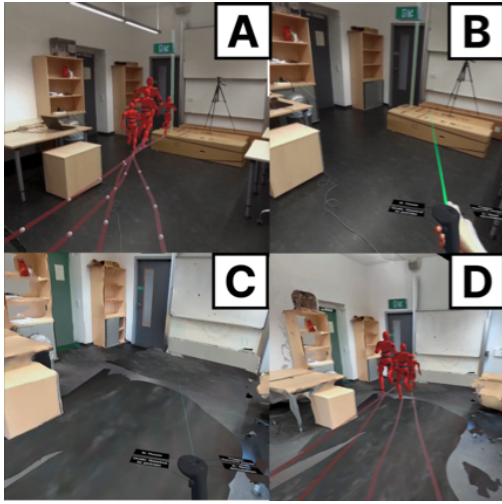


Fig. 7: In Scenario 1, (A) the user noticed agents had to detour around boxes during escape. (B-C) The user then tried to remove the box in AR, and (D) via rerunning the simulation, discovered it reduced detour.

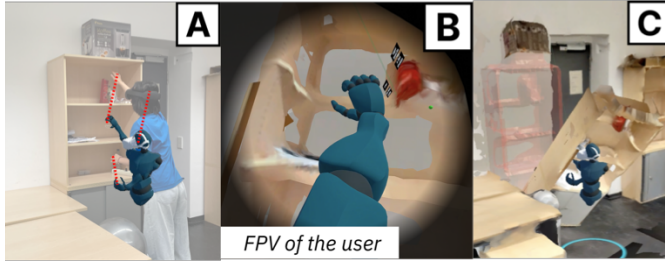


Fig. 8: Part of Scenario 2. (AB)The user embodied a kid and tried to reach the shelf, and (C)found that the shelf might tip over.

the additional implementation. We offer a **a video figure** to showcase scenarios.

4.1 Scenario 1: Fire Evacuation Planning

Background. Tom, the campus safety officer, needed to check the fire security of a lab that usually accommodated four students. The lab had two doors, one being an emergency exit.

Motivation. With ESS, Tom aimed to evaluate the evacuation efficiency of the current spatial layout and identify hidden dangers.

Procedure. As indicated in Fig.1, Tom identified the hidden danger caused by the misposition of fire alarm and visual ability of occupants. He then hoped to decrease the evacuation time. He analyzed the trajectory and found that agents had to detour around boxes while escaping (Fig.7A). Unable to move the heavy boxes immediately, he virtually removed them in AR and re-run the simulation (Fig.7BC), then he found that the elapsed time decreased (Fig.7D). For safety, he decided to remove the boxes.

4.2 Scenario 2: Curating VR Exhibition for Children

Background. Sam would host a VR exhibition for fifth-grade students at the campus VR lab. The kids would explore VR demos and the lab.

Motivation. With ESS, Sam aimed to ensure kids' safety and smooth traffic in the lab.

Procedure. Sam loaded five agents in the height of fifth-grade students in the room. He created a virtual desk for placing VR devices, then simulated agents' walking behavior to test how the desk's position influence traffic flow. Then he simulated kids' motions when playing a VR boxing game. It showed that the kid's wild movements caused accidental hits to others. So Sam decided to change the VR game. Additionally, in the simulation, Sam observed that children gathered around a shelf showing interest in devices placed there. He thought the

children might grab things on the shelf. So he embodied a child's body, stood on tiptoe and tried to reach the items (Fig.8AB). Suddenly, the shelf tipped over (Fig.8C). He realized the shelf was top-heavy due to a box on top, making it prone to tipping if a short kid interacted with it. This reminded him of the IKEA dresser issue. To prevent accidents, Sam removed items from the shelf and decided to anchor the shelf to the wall.

4.3 Customised implementation for scenarios

All interactions in the scenarios were implemented and could be experienced sequentially according to the described storyline. We expanded the ESS prototype with extra functions tailored to the scenario stories. Firstly, we simulated color blindness in Scenario 1. We altered the RGB value of the 3D reconstruction texture. Additionally, we enabled users to virtually manipulate real-world objects in Scenario 1 and 2. Examples are shown in Fig.7 and Fig.8. To implement, we separated individual objects off 3D reconstruction mesh using Blender3D³ and made them manipulatable (e.g., rotate, remove) in Unity3D.

These functionalities could be further developed. For instance, adjusting the AR render setup could simulate various visual impairments [37]. We only implemented features customized for our scenarios, because our focus was on including the essential elements to effectively demonstrate the concept.

5 PILOT STUDY: SCENARIO WALKTHROUGH WITH XR RESEARCHERS

In our pilot study, we invited 4 XR researchers for scenario walkthroughs by using our prototype [42]. The goal was to get feedback on the concept, core functionalities and overall experience. We chose XR researchers, as they can assess user experience and concept beyond the current mid-fidelity AR prototype, which focused on core functions over algorithm depth.

5.1 Participants

We recruited four XR researchers (M=4, aged between 23 to 40, E1-E4) to participate in our study. All of them had experience (3 to 6 years) in XR research and development. We followed the institutional ethics regulations. All participants provided informed consent and were informed that they could withdraw from the study at any time. All data were anonymized for analysis.

5.2 Study Procedure

The study was conducted in our lab. One researcher moderated the session. Each study session lasted around one hour and consisted of three phases. Firstly, we introduced the concept and study tasks, then invited participants to sign consent forms. Secondly, participants put on the HMD and went through the two scenarios under the guidance of the researcher. Participants were encouraged to think aloud. Lastly, we conducted a semi-structured interview to understand participants' experience of using the prototype and their general impression on this concept.

5.3 Data Analysis

We collected and transcribed the audio recording of think-aloud remarks and interviews. Two authors conducted an open-coding analysis [33] on the transcription. It started with the first author reading through the quotes and developing an initial code book. Then, the two authors categorized the quotes into themes through discussion.

5.4 Results

In this section, we presented our findings as the following themes extracted from the open-coding analysis.

Agent embodiment unleashes new experience, thoughts and emotions. Participants highlighted embodying different agents offered new experiences, insights, and emotions in decision-making. E2 highlighted the novel experience of embodying a small person contrasting to reality, "I changed to a one-meter-high kid, so it is not normal to see

³<https://www.blender.org/>

and run around ... That was quite nice because I'm quite tall." Such novel experience could bring new thoughts and potentially change decisions, "Getting this perspective truly helped, I wouldn't have done [the reposition of the fire alarm] if I had done it from my own perspectives" [E4]. Through embodiment participants began to consider different individuals when making decisions, "I would embody people with just one arm to interact with the world. Then you might design things differently, for example, the way you open the door" [E4]. In addition, Agent Embodiment allows participants to experience different emotions, "I could actually kind of feel helpless [when embodying a small agent]" [E4].

Agent embodiment shifts focus to problems and reasoning behind simulated behavior. Participants thought that Agent Embodiment helped them to discover problems and understand the reasoning behind the simulation results. All the participants mentioned that changing perspectives helped them to discover negligible issues. For example, E3 noted that embodying the agent in situ nudged him to pay closer attention to details and uncover more hidden dangers., "When I'm just having an overview, I feel quite relaxed about the card boxes (in Scenario 1), people should walk around it easily. But when I embodied it, I felt that this was not well visible and the chance of tripping over it is way higher." E4 also mentioned, "When you make decisions for many people, it is easy to overlook something. But if you can see when they see, for example, color blindness, you might realize that their experience is not good." Additionally, direct experience could help people better understand the reasoning behind the simulated behavior. For instance, in the fire evacuation scenario, embodying colorblind agents truly helped participants to understand how visual impairments could complicate navigation and lead to mistakes, "It's hard to get to this point at first thought. I've never thought about this before taking their perspective" [E4].

Improvement Opportunities for the Prototype. We identified opportunities to improve the prototype. First, three participants (E2–E4) noted that the low fidelity of virtual humans could undermine the user experience for lay users. E3 suggested that behavioral fidelity might affect users' trust in the simulations: "It is too pixelated. I'm not sure whether people would trust a situation like this." Second, scalability was raised by E2, who suggested combining AR with an HMS desktop application to support both individual perspectives and large-scale crowd management: "The desktop app would be sufficient to see all people moving around, while AR could still provide the perspective of a child."

Future Scenarios for ESS. Participants mentioned a variety of everyday contexts where ESS could be applied. For example, E1 mentioned: "If you know that you have a friend in a wheelchair and you want to set up the buffets and make sure that they can reach everything". All participants also noted scenarios requiring awareness of diverse populations. E2 also said, "...at a train station... I can simulate how they walk and look, or how high I have to set up this sign and whether I can still read it?"

In general, experts recognized the value of concept. They highlighted the potential benefits of Agent Embodiment to offer new perspectives and shift focus to the reasoning behind the simulated behavior. We also gathered feedback on the prototype for improvement.

6 ONLINE SURVEY STUDIES WITH VARIOUS OCCUPATIONS

Based on the feedback from expert walkthroughs, we recognized the potential of Agent Embodiment to shift people's focus to understanding the reasoning behind the simulated behavior. Rather than measuring its physiological impacts, we were curious whether this conceptual effect remains present and valuable within a more diverse population. We conducted an online survey with people in different professions. The study goal was to collect diverse feedback on the overall perception and perceived usefulness of this concept, as well as to understand how people would incorporate ESS into their professional or daily life scenarios.

Public Safety	Fireman (P13), Police (P14, P16, P19), Prison Officer (P31)
Healthcare	Nurse (P1, P7, P8, P11, P15), Dentist (P4), Social Care Worker (P17), Mental Health Supporter (P33)
Design	Media Buyer (P3), Graphic Designer (P5), Freelance Artist (P30, P32)
Architecture and Mechanics	Architect (P18), Civil Engineer (P21), Logistic Manager (P23), Emergency Technical Engineer (P29), Operations Specialist (P35)
Education	Teacher (P6, P9, P24, P27), Elementary School Principal (P26)
Human Communication	Customer Experience Specialist (P12), Human Resource Manager (P22), Bookseller (P25), Personal Assistant (P37), Marketing Specialist (P28, P38)
Other	Accountant (P2), Janitor (P20), Software Developer (P34, P36), Manufacture Worker (P10)

Table 1: Participants in survey study

6.1 Method

We used an online survey to reach participants from various backgrounds. Demonstrating the concept through videos also helped participants focus on the concept rather than the fidelity limitations of AR prototypes, which often distract non-expert users when using XR applications [67]. We followed the methods used by Rixen et al. and Koelle et al. to evaluate novel AR concepts through online surveys. This approach involved presenting scenario videos along with questions to capture participants' opinions [34, 65, 66]. Krauss et al. emphasized the advantages of using storytelling videos for concept communication in XR development [36], as they allow users to focus on the concept itself rather than technical performance, particularly for lay users without prior experience with proof-of-concept prototypes.

To understand the value of In-situ Simulation and Agent Embodiment, we presented the concept in two steps. First, we introduced In-situ Simulation without the embodiment concept. We showed a video of an AR user simulating agents' navigation behavior in different room layouts. Participants then (1) evaluated the usefulness of In-situ Simulation with the *Perceived Usefulness Questionnaire* (PUQ) [13], (2) rate their willingness to use this technology ("Q: Overall, if the technology is accessible now, I will adopt this technology in my work"), and (3) generated scenarios by integrating it into their daily work. After that, we introduced the full ESS concept, "In-situ Simulation with Agent Embodiment". We explained embodiment with illustrations. We then presented a second video showing an AR user embodying the agent from the previous video. This agent was small and colorblind, and the user directly experienced the agent's walking process. Participants were then asked to evaluate the full concept with the same questionnaire and scenario generation tasks.

6.2 Participants

We recruited participants from *Prolific*. The screening criteria included: (1) fluency in English, (2) basic knowledge and use experience of AR. We included both office and field-based professions, which involves interacting with space or understanding people, including public safety (e.g., police) and space planning (e.g., architects), as well as four other domains, to get diverse feedback. Details are shown in Table 1. Participants provided informed consent and were informed of the study's purpose, as well as their right to withdraw at any time. All data were anonymized for analysis.

6.3 Procedure

The survey consisted of 5 parts:

Part 1: Daily job and task We asked about participants' occupations, daily tasks, and challenges. The goal is to sensitize participants and help them connect the concept with their daily tasks.

Part 2: Partial Concept Presentation We introduced In-situ Simulation without embodiment, as illustrated in Section 6.1. Participants were asked to watch the concept video, answer video-checking questions, fill in the questionnaires, and generate scenarios by integrating the concept into their daily work.

Part 3: Full Concept Presentation We introduced the full concept ‘In-situ Simulation with Agent Embodiment’, following the same process and tasks as in Part 2.

Part 4: Generate More Scenarios We presented 2 scenarios with video as inspiration, inviting participants to create scenarios by relating ESS to their personal lives.

Part 5: Wrap-up Question We posed open questions asking participants to consider (1) additional desirable features, (2) potential concerns, and (3) overall opinions on the concept.

For each video, we included checking questions to confirm participants watched the video and understood the concept. The PUQ questionnaire included attention-checking questions to ensure participant engagement. The survey took about 30 minutes, with each participant receiving £5.

6.4 Data Collection and Analysis

We recruited 38 participants (19 male, 18 female, 1 non-binary) aged 21 to 60 ($M=31$, $SD=9.28$). Of 57 initial responses, 38 were retained after filtering. We excluded responses completed in under 15 minutes and only accepted those that passed all video-understanding and attention-checking questions. Both qualitative and quantitative data were collected.

For the quantitative data, we collected responses of PUQ and “willingness to adopt the technology” for both “In-situ Simulation” and “In-situ Simulation with Agent Embodiment.” The PUQ uses a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree), where higher scores reflect greater perceived usefulness. We averaged the six items from each response to derive an overall PUQ score. We also assessed willingness to adopt the technology with one 7-point Likert scale question (“Q: Overall, if the technology is accessible now, I will adopt this technology in my work”), where higher scores indicate stronger intentions.

For qualitative data, we collected responses in the open-ended questions. We performed open-coding analysis following the same process as in Sec.5.3.

6.5 Quantitative Findings

Perceived Usefulness The PUQ results showed In-situ Simulation ($M = 5.49$, $SD = 1.06$) and “In-situ Simulation with Agent Embodiment” ($M = 5.38$, $SD = 1.40$). For both measures, the mean scores are above 5, indicating a “More or less agree” response. This suggests that participants generally perceive the concept as useful in their daily work context.

Willingness to adopt the technology The result of the willingness in technology adoption showed In-situ Simulation ($M = 5.66$, $SD = 1.27$) and “In-situ Simulation with Agent Embodiment” ($M = 5.61$, $SD = 1.46$). For both, their mean scores are above 5 (More or less agree), indicating that participants are generally open to the possibility of adopting this technology for future job tasks.

6.6 Qualitative Findings

We analyzed two sets of scenarios that participants respectively generated with In-situ Simulation and “Simulation with Agent Embodiment” in their job contexts. We used a hybrid inductive–deductive approach. We first created categories by reading part of the data. We then applied these categories to the rest of the data, while still adding or combining codes as needed. We used Excel to code the data for easily organizing and filter it. Two authors analyzed the data together using a large touchscreen. In the initial coding session, the first author read the data and created the code book. It was found that most codes fell into two patterns: (1) specific problems to solve and (2) big-picture benefits. Then two authors synthesized these codes into two themes: *task* and *value*. The *task* theme refers to how people want to use the technology and the tasks they aim to achieve. The *value* theme referred to the overall benefits that people sought to bring to the final decision through using this technology. Disagreements were resolved through discussion. We offer our codebook in the supplementary material.

6.6.1 In-situ Simulation: Task

We analyzed the tasks people want to achieve through using In-situ Simulation. Participants responded whose and what behavior they wanted to simulate. Additionally, participants also mentioned the solutions they aimed to obtain by simulating future conditions. We divided them into three classes:

(1) Forseeing an individual's behavior to prepare own reaction.

This theme was noted by 11 participants, with relevant tasks involving significant social and physical interactions, such as teaching, nursing, and policing. Simulation revolved around the unique behavior of a specific individual in a small-scale social context. The goal is to predict a specific person's action and then provide an accurate and well-prepared response. People aimed to address potential safety concerns, financial losses, and thoughtless interpersonal interactions that could result from an unprepared and rushed reaction. For example, P1, a nurse, mentioned: “*Being able to see how a patient will respond to needles would be revolutionary [...]*” P14, a police officer, discussed simulating a perpetrator's sudden attack to help police prepare for self-defense: “*Seemingly out of nowhere, he became violent towards officers which resulted in one officer getting assaulted. [...] If these glasses (headset) existed, then officers could have assessed the situation and the [man]'s body language.*”

(2) Simulating happenings to optimize environment or event management.

Twenty participants emphasized the importance of simulating events, including environmental factors, human interactions, and specific occurrences, as a key tool for informing decisions related to crowd planning, spatial design, and event management. The goal was to enhance spatial efficiency, public safety, or inclusiveness. Participants sought to refine a decision or select the best option by testing it in specific scenarios. The tasks could be simulating human interactions to evaluate environmental factors, aiding spatial design. For example, P17, a social care worker, mentioned: “*[...] plan the size of corridors and doorways to ensure they can accommodate wheelchairs.*” P27, a teacher, also noted: “*how certain sitting arrangements impact my students.*” The tasks could also involve simulating group behavior in a particular context and guiding event management. P14, a policewoman, mentioned simulating group behavior of bar fights to decide on policing strategy, “*fights in crowded spaces, such as pubs or bars [...]. With the headset, we could evaluate the behavior of the room and assess where is likely to cause the most issues so that officers can react accordingly.*”

(3) Simulating own future behavior to inform next step.

Four participants mentioned how watching their own simulated behavior of their future selves can plan or optimize their next-step actions in different tasks. These tasks could involve physical activities to enhance time and effort efficiency, as P21, a janitor, noted: “*It could help me see where I can reach to clean heavy items before you attempt to move them.*” This approach could also involve social interactions to evaluate one's communication strategy. For instance, P28 aimed to simulate and observe her teaching behavior to inform interaction with students: “*It would enable me to evaluate how different teaching approaches or strategies are effective on my students.*”

6.6.2 In-situ Simulation: Value

Participants identified efficiency, safety, and preparedness as the key benefits that this concept can bring to their decision-making.

(1) Efficiency. Participants mentioned simulating various future conditions to see the required time and effort for choosing the most resource-efficient option. The goal is to improve efficiency for future actions. For instance, “*it can help me with a route to take to minimize time to get the patient to intensive care in a moment (P8).*” This code was the most cited benefit, mentioned by 14 participants.

(2) Safety with Foresee and Proaction. 10 participants emphasized the value of using this technology in safety-related contexts, primarily for predicting and subsequently avoiding or preventing potential hazards, such as, “*It would help me foresee issues a child will have had in my previous care roles (P37),*” and “*something they could trip over can be easily identified (P9).*”

(3) Positive Emotion with Preparedness. Three participants mentioned that human simulation offers emotional support by improving

preparedness for unexpected happenings, with keywords: “*calm* (P7)” and “*confident* (P19).” For instance, P1, a nurse, discussed simulating patients’ reactions to injection for her preparation: “*[To simulate] how [patients] will respond to needles would be revolutionary. It would take away a lot of the stress and psychological load.*”

6.6.3 Simulation with Agent Embodiment: Task

Through generating scenarios, participants described whom they wished to simulate and embody, and what experiences they aimed to understand. Unlike In-situ Simulation, in most of the scenarios, participants did not specifically expect an actionable outcome as the solution. Instead, they focused on gaining insights by experiencing specific behaviors and then using those insights to frame decisions. They mainly described their motivation as understanding or reflecting through experiencing, as noted by participants “*great to feel*” [P1], “*interesting to see how different*” [P19], “*make a lot of food for thoughts*” [P25], “*to understand their challenges*” [P27]. We categorized them into five classes:

(1) Go through others’ holistic experience for understanding. Fourteen participants mentioned about holistic experience. It is defined as a high level of experience including emotional, cognitive, and physical experience. Examples included therapy experience (P1, P4), the sensation of suicidal individuals (P14), the experience of victims in domestic violence (P19), the experience leading to crime commission (P38), interpersonal communication experiences (P17, P22, P38), and the experience of being another gender (P19). As P1, a nurse, sought to embody patients undergoing therapy to enhance their well-being, “*It would be great to feel how every patient experiences their therapy. As a nurse we rarely get to see the ‘other side’ so this would be a game changer.*”

(2) Experience others’ bodily ability to optimize the environment. Sixteen participants expressed a desire to explore how people physically interact with their environment. They mainly aimed to enhance ergonomic efficiency and physical accessibility of a physical environment. For instance, P5 mentioned: “*I could take the role of a person in a wheelchair to check if they have easy access to all the office areas.*”

(3) Experience others’ physical sensations for understanding. This theme was noted by 12 participants, focusing on how people perceive the world through various sensations, including pain (P1, P15), sight impairment (P6, P25), hearing loss, touch and light sensitivity (P24). P1, a nurse, aims to improve the interaction with patients during injection, mentioned: “*I would love to experience the pain sensitivity everyone else around me has. I would consider myself as highly tolerant to pain. But I see people wince at almost anything nowadays.*”

(4) Experience others’ cognitive ability for understanding. Seven participants mentioned their motivation to embody and understand people’s cognitive capabilities, such as mental calculation (P2), dyslexia (P6), cognitive overload under stress (P8, P12, P22), and limited ability to retain knowledge (P27). P6, a teacher, hoped to improve teaching strategy for students with dyslexia, “*[Through embodiment] I could ‘feel’ how a dyslexic student perceives text and numbers.*”

(5) Experience others’ perceptions of oneself for self-reflection. Two participants mentioned embodying others and viewing themselves through the eyes of others. The goal is to support self-reflection in social contexts. For example, P25, a bookseller, mentioned embodying customers to enhance business communication: “*I could see how customers see me and my colleague, that would make a lot of food for thoughts.*”

6.6.4 Simulation with Agent Embodiment: Value

Participants emphasized that they hope this technology can bring individualization, empathy, and comfort to their decisions.

(1) Individualisation. Thirteen participants emphasized its importance for understanding an individual’s unique needs or situations, which helps in customizing solutions and making decisions more precise. They highlighted the subtle differences between individuals could only be understood by experiencing them. For example, P15 mentioned mental health, “*Mental health is so personal and different from person to person and being able to feel the sensations of someone’s feelings*

would allow me to be more connected to them and know how to help them as best as I can.”

(2) Empathy. Twelve participants emphasized the motivation to build emotional connections through genuine concern for others’ well-being, which supports better communication when implementing decisions. For example, P19, a police, mentioned his experience in communicating with victims: “*Putting me in the place of a victim, such as a female in a violent domestic abuse position, would help improve my empathetic response in real life situations.*”

(3) Subtle Comfort. Seven participants highlighted that comfort, which is too nuanced to anticipate through mathematical metrics, could only be understood through direct experience. This includes social comfort, such as “*[through embodiment I want to be] able to understand how people around me perceive me, being able to understand if I make them feel uncomfortable or if they like my presence* (P4).” Three participants also highlighted physical comfort, which went beyond ergonomic efficiency. P21, a civil engineer, mentioned: “*Maybe [to embody] a very tall person to feel how they feel having to lower their heads trying to get past the door... what adjustments can be done to improve their comfort.*”

7 DISCUSSION, LIMITATION, AND FUTURE WORK

Through two studies, we reflected on (1) AR broadening HMS usage, (2) experiencing simulated behavior might shift focus from following the simulation results to reflecting on them, and (3) expanding ESS to other embodied experiences.

7.1 AR’s potential to facilitate wider adoption of HMS in diverse contexts

AR is increasingly prevalent, offering people new ways to interact with data in relevant real-world contexts [17]. We anticipate that AR will also facilitate a new way to interact with HMS, leading to broad and creative applications across various scenarios. This vision contrasts with the current situation, where HMS is primarily utilized by experts in specific fields like mechanics [1].

Our survey results revealed this potential, in which we invited potential everyday users from various domains to envision using ESS in their own lives. We gathered diverse scenarios from the survey, from supporting teaching interaction and preparing injections for patients, to guiding cleaning tasks. Additionally, the questionnaire responses indicated that participants perceived this concept as useful and expressed strong motivation to adopt it. For some of the collected scenarios, people might not associate them with HMS. They seem to be too “trivial” to use HMS tools. Currently, HMS tools are mostly desktop software that requires professionals to spend hours sitting at a computer to configure parameters [1, 40]. This situation is due to current limitations in technology performance and interaction design, but it does not imply that HMS should be restricted to professional use by experts alone. We expect a future where HMS can quickly and accurately simulate different aspects of human behavior [16, 22, 38, 63]. With AR allowing people to directly see and experience simulated behavior, HMS would be used by everyone for different tasks and in diverse contexts. One good example is how photography technology became accessible to everyone. In the past, taking photos required a professional with complex equipment and a long wait for film development. People only took photos for once-in-a-lifetime events [9]. Now with mobile phones, anyone can capture images everywhere with just a click. This shift sparked a lot of creative uses for photography, from social media to crowdsourcing urban data [28, 54]. Recently we are seeing a similar trend with large language models, which are used in many different ways that we might not have imagined before [25]. Therefore, we anticipate a similar future for HMS, with widespread adoption for various creative applications. However, the wider adoption of HMS still relies on future algorithm efficiency, design, and ethical concerns like personal behavior data privacy [51]. Without these, broader use may remain out of reach.

7.2 Direct Experience as a sort of “Explainability” to Understand Simulation Results

From the study, we found that ESS can shift people's focus from simply relying on simulation outcomes for the final solution to reflecting on the experiences behind simulated behavior for deeper insights. We hope this shift could encourage users to combine their own insights with the simulation model, leading to more thoughtful and human-centered decision-making.

We see the potential from the survey results. In the generated scenarios with In-situ Simulation, participants mainly expected a practical outcome from the simulation results and would take that as the final decision. They expected the simulation to answer *what to do* and *how to do it*, such as identifying the shortest path or most effortless cleaning plan. However, when generating scenarios with the full concept of ESS, participants focus more on embodying others to understand their experiences and perspectives behind their behavior. Experiencing could be the goal in itself. It helps to understand and reflect on the reasoning or underlying factors (e.g., social comfort) behind a behavior [59, 69]. Additionally, as mentioned in Sec 6.6.2, participants prioritized the “Efficiency”, “Forsee and Proaction”, “Preparedness” as the key values when using In-situ Simulation to inform decisions. In contrast, with the full ESS concept, participants place more emphasis on *Individualization*, *Empathy*, and *Subtle Comfort* in their decisions (Sec 6.6.4). These nuanced factors are better understood and addressed through human judgment rather than simulation models [43, 50]. By *thinking through experiencing*, people might combine their own knowledge and intuition to make decisions, rather than relying solely on the simulated results. As P14, a police officer, explained her motivation for experiencing the simulated behavior of victims: “*This could help me think about what type of comfort they are needing in that moment.*”

Understanding the process behind the machine intelligence outcomes is not unique to this case. Recently, Explainable Artificial Intelligence (XAI) is a popular topic, which also aims at clarifying the thinking process behind the AI prediction results. Current XAI methods use data visualization and textual descriptions to explain the underlying mechanisms of a model's behavior [6]. **We view ESS as a form of XAI that uses a different approach – direct experience – to help people understand the experience or reasoning behind simulated behaviors.** Furthermore, we hope ESS could allow individuals to independently reflect on and interpret the simulation results, and use their own insights to adapt the decisions proposed by the model. This is valuable in complex or nuanced situations where purely data-driven decisions might miss the subtleties that human judgment can capture.

7.3 Extend ESS with more embodied experience

Based on our survey results, more types of embodied experiences could be simulated and transformed for users to optimize ESS. From the generated scenarios, participants expressed interest in a wider range of experiences than the example scenarios we provided. They were interested in directly experiencing another one's physical sensations, mental experience, and cognitive processes, such as pain, depression, or dyslexia. Some of these experiences are possible with the current state of the art. For physical sensations, sensory techniques have been developed to replicate senses like hearing [24], touch [55], pain [7], and limb motion [52, 57]. For cognitive capabilities, prior research explores using visual augmentation techniques to simulate reading difficulties of dyslexic individuals [70]. With the ESS concept, we primarily focus on implementing visual perceptions and physical capabilities to demonstrate our vision. However, we believe our proof of concept can be extended to cover a greater variety of embodied experiences. This would help people understand the more specific needs of diverse individuals.

7.4 Limitation and Future Work

Scalability. We are aware that ESS might not fit the typical HMS tasks like crowd planning, which require scalability. AR may not provide environment overview crucial for crowd management. However, this work focuses understanding specific individuals' needs in smaller contexts. This focus aligns with our survey results, where participants

expressed interest in using ESS in similar scenarios. We expect a combined approach using desktop simulations for large-scale environments, complemented by ESS to simulate the experiences of specific populations in targeted areas.

Technical Depth of Prototype. We realize the technical limitations of our implementation, which used simple algorithms (e.g., NavMesh) to simulate navigation for concept demonstration. It may limit the context richness and decision-making depth. Future work could integrate real-world simulation models [61, 73], incorporating complex behavioral factors and decision-making contexts.

Diversity of Participants in Study 1. We acknowledge that the demographic diversity of participants in Study 1 was limited, as all participants were male XR researchers, which this study primarily focused on participants with XR expertise to evaluate an early-stage prototype. In future work, we plan to recruit a more diverse group of participants in terms of gender, age, and professional background to elicit broader perspectives on how this concept can be used and improved.

Method of Study 2. In Study 2, we used an online survey with video demonstrations for conceptual evaluation. While we chose this method to encourage lay users to focus on the concept rather than technology fidelity, we acknowledge that it lacks hands-on XR experience. So the feedback related to practical usage (e.g., usability, comfort) might not be captured. As a next step, we plan to conduct on-site studies with an improved prototype intended for real-world use.

8 CONCLUSION

We presented the concept of Embodied Spatial Simulation (ESS), allowing an XR user to configure “what-if” scenarios in-situ, simulate motion behavior of virtual agents, and then embody these agents to experience the world from their perspectives. We explored the concept with an XR prototype applied to two scenarios. Through scenario walk-throughs (N=4) and an online survey (N=38), we found that ESS had the potential to let people focus more on reflecting on the experience behind a simulated behavior, rather than following the simulation results to make decisions. We also collected diverse scenarios to use ESS, where HMS was not used traditionally. We aim to offer a new perspective that emphasizes “experiencing the simulation results” instead of merely executing the suggested decisions offered by machine intelligence.

ACKNOWLEDGMENTS

This work has been funded by the German Federal Ministry of Education and Research and the Hessian Ministry of Higher Education, Research, Science and the Arts within their joint support of the National Research Center for Applied Cybersecurity ATHENE and by the LOEWE initiative (Hesse, Germany) within the emergenCITY centre.

REFERENCES

- [1] K. Abdel-Malek and J. S. Arora. *Human motion simulation: predictive dynamics*. Academic Press, 2013. 1, 2, 8
- [2] H. S. Alavi, H. Verma, J. Mlynar, and D. Lalanne. The hide and seek of workspace: Towards human-centric sustainable architecture. In *Proceedings of the 2018 CHI conference on human factors in computing systems*, pp. 1–12, 2018. 2
- [3] H. S. Alavi, H. Verma, M. Papinutto, and D. Lalanne. Comfort: A coordinate of user experience in interactive built environments. In *Human-Computer Interaction-INTERACT 2017: 16th IFIP TC 13 International Conference, Mumbai, India, September 25–29, 2017, Proceedings, Part III 16*, pp. 247–257. Springer, 2017. 2
- [4] B. Alenljung, J. Lindblom, R. Andreasson, and T. Ziemke. User experience in social human-robot interaction. In *Rapid automation: Concepts, methodologies, tools, and applications*, pp. 1468–1490. IGI Global, 2019. 2
- [5] J. Amores, X. Benavides, and P. Maes. Showme: A remote collaboration system that supports immersive gestural communication. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*, pp. 1343–1348, 2015. 3

- [6] A. B. Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. García, S. Gil-López, D. Molina, R. Benjamins, et al. Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai. *Information fusion*, 58:82–115, 2020. 9
- [7] C. Asada, K. Tsutsumi, Y. Tamura, N. Hara, W. Omori, Y. Otsuka, and K. Sato. Electrical muscle stimulation to develop and implement menstrual simulator system. *Journal of Robotics and Mechatronics*, 33(5):1051–1062, 2021. 9
- [8] D. Banakou, R. Groten, and M. Slater. Illusory ownership of a virtual child body causes overestimation of object sizes and implicit attitude changes. *Proceedings of the National Academy of Sciences*, 110(31):12846–12851, 2013. 3
- [9] W. Benjamin. A short history of photography. *Screen*, 13(1):5–26, 1972. 8
- [10] W. Büschel, A. Lehmann, and R. Dachzelt. Miria: A mixed reality toolkit for the in-situ visualization and analysis of spatio-temporal interaction data. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1–15, 2021. 2
- [11] D. B. Chaffin. Human motion simulation for vehicle and workplace design. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 17(5):475–484, 2007. 1, 2
- [12] T. Chandler, M. Cordeil, T. Czauderna, T. Dwyer, J. Glowacki, C. Goncu, M. Klapperstueck, K. Klein, K. Marriott, F. Schreiber, et al. Immersive analytics. In *2015 Big Data Visual Analytics (BDVA)*, pp. 1–8. IEEE, 2015. 2
- [13] F. D. Davis. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS quarterly*, pp. 319–340, 1989. 6
- [14] P. Dourish. Embodied interaction: Exploring the foundations of a new approach to hci. *Work*, pp. 1–16, 1999. 3
- [15] P. Dourish. *Where the action is: the foundations of embodied interaction*. MIT press, 2001. 3
- [16] N. A. ElSayed, B. H. Thomas, K. Marriott, J. Piantadosi, and R. T. Smith. Situated analytics: Demonstrating immersive analytical tools with augmented reality. *Journal of Visual Languages & Computing*, 36:13–23, 2016. 2, 8
- [17] B. Ens, B. Bach, M. Cordeil, U. Engelke, M. Serrano, W. Willett, A. Prouzeau, C. Anthes, W. Büschel, C. Dunne, et al. Grand challenges in immersive analytics. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1–17, 2021. 2, 8
- [18] T. Feng, L.-F. Yu, S.-K. Yeung, K. Yin, and K. Zhou. Crowd-driven mid-scale layout design. *ACM Trans. Graph.*, 35(4):132–1, 2016. 1, 2
- [19] D. Freeman, N. Evans, R. Lister, A. Antley, G. Dunn, and M. Slater. Height, social comparison, and paranoia: An immersive virtual reality experimental study. *Psychiatry research*, 218(3):348–352, 2014. 3
- [20] G. Freeman and D. Acena. "acting out" queer identity: The embodied visibility in social virtual reality. *Proceedings of the ACM on Human-Computer Interaction*, 6(CSCW2):1–32, 2022. 3
- [21] N. Ganganath, C.-T. Cheng, T. Fernando, H. H. Iu, and K. T. Chi. Shortest path planning for energy-constrained mobile platforms navigating on uneven terrains. *IEEE Transactions on Industrial Informatics*, 14(9):4264–4272, 2018. 2
- [22] M. Gillies and N. A. Dodgson. Behaviourally rich actions for user-controlled characters. *Computers & Graphics*, 28(6):945–954, 2004. 8
- [23] M. Gonzalez-Franco and T. C. Peck. Avatar embodiment. towards a standardized questionnaire. *Frontiers in Robotics and AI*, 5:74, 2018. 3
- [24] N. Grimault, T. Irino, S. Dimachki, A. Corneillie, R. D. Patterson, and S. Garcia. A real time hearing loss simulator. *Acta Acustica united with Acustica*, 104(5):904–908, 2018. 9
- [25] M. U. Hadi, R. Qureshi, A. Shah, M. Irfan, A. Zafar, M. B. Shaikh, N. Akhtar, J. Wu, S. Mirjalili, et al. A survey on large language models: Applications, challenges, limitations, and practical usage. *Authorea Preprints*, 2023. 8
- [26] C. Hurter, N. H. Riche, S. M. Drucker, M. Cordeil, R. Alligier, and R. Vuilleumot. Fiberlay: Sculpting three dimensional trajectories to reveal structural insights. *IEEE transactions on visualization and computer graphics*, 25(1):704–714, 2018. 2
- [27] C. Javerliat, S. Villenave, P. Raimbaud, and G. Lavoué. Plume: Record, replay, analyze and share user behavior in 6dof xr experiences. *IEEE Transactions on Visualization and Computer Graphics*, 2024. 2
- [28] N. Jurgenson. *The social photo: On photography and social media*. Verso Books, 2019. 8
- [29] M. Kari, R. Schütte, and R. Sodhi. Scene responsiveness for visuotactile illusions in mixed reality. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–15, 2023. 3
- [30] S. Kasahara, M. Ando, K. Suganuma, and J. Rekimoto. Parallel eyes: Exploring human capability and behaviors with paralleled first person view sharing. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pp. 1561–1572, 2016. 3
- [31] S. Kasahara and J. Rekimoto. Jackin: integrating first-person view with out-of-body vision generation for human-human augmentation. In *Proceedings of the 5th augmented human international conference*, pp. 1–8, 2014. 3
- [32] S. Kasahara and J. Rekimoto. Jackin head: immersive visual telepresence system with omnidirectional wearable camera for remote collaboration. In *Proceedings of the 21st ACM symposium on virtual reality software and technology*, pp. 217–225, 2015. 3
- [33] S. H. Khandkar. Open coding. *University of Calgary*, 23(2009), 2009. 5
- [34] M. Koelle, S. Ananthanarayan, and S. Boll. Social acceptability in hci: A survey of methods, measures, and design strategies. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1–19, 2020. 6
- [35] M. Kraus, K. Klein, J. Fuchs, D. A. Keim, F. Schreiber, and M. Sedlmair. The value of immersive visualization. *IEEE computer graphics and applications*, 41(4):125–132, 2021. 2
- [36] V. Krauß, M. Nebeling, F. Jasche, and A. Boden. Elements of xr prototyping: Characterizing the role and use of prototypes in augmented and virtual reality design. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pp. 1–18, 2022. 6
- [37] K. Krösl, C. Elvezio, L. R. Luidolt, M. Hürbe, S. Karst, S. Feiner, and M. Wimmer. Cataract: Simulating cataracts in augmented reality. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 682–693. IEEE, 2020. 3, 5
- [38] J. J. Kuffner and J.-C. Latombe. Fast synthetic vision, memory, and learning models for virtual humans. In *Proceedings Computer Animation 1999*, pp. 118–127. IEEE, 1999. 8
- [39] B. Lee, T. Jin, S.-H. Lee, and D. Saakes. Smartmanikin: virtual humans with agency for design tools. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2019. 1, 2, 3
- [40] B. Lee, M. Lee, J. Mogk, R. Goldstein, J. Bibliowicz, F. Brudy, and A. Tessier. Designing a multi-agent occupant simulation system to support facility planning and analysis for covid-19. In *Designing interactive systems conference 2021*, pp. 15–30, 2021. 2, 3, 8
- [41] B. Lee, M. Lee, P. Zhang, A. Tessier, D. Saakes, and A. Khan. Socio-spatial comfort: Using vision-based analysis to inform user-centred human-building interactions. *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW3):1–33, 2021. 2
- [42] C. Lewis and C. Wharton. Cognitive walkthroughs. In *Handbook of human-computer interaction*, pp. 717–732. Elsevier, 1997. 5
- [43] J. Li, H. Cao, L. Lin, Y. Hou, R. Zhu, and A. El Ali. User experience design professionals' perceptions of generative artificial intelligence. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 1–18, 2024. 9
- [44] X. Li, X. Tang, X. Tong, R. Patibanda, F. Mueller, and H.-N. Liang. Myopic bike and say hi: games for empathizing with the myopic. In *Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play*, pp. 333–338, 2021. 3
- [45] L. Lin, A. Normoyle, A. Adkins, Y. Sun, A. Robb, Y. Ye, M. Di Luca, and S. Jörg. The effect of hand size and interaction modality on the virtual hand illusion. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 510–518. IEEE, 2019. 3
- [46] D. Lindlbauer and A. D. Wilson. Remixed reality: manipulating space and time in augmented reality. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2018. 3
- [47] S. A. Linkenauger, H. H. Bühlhoff, and B. J. Mohler. Virtual arms reach influences perceived distances but only after experience reaching. *Neuropsychologia*, 70:393–401, 2015. 2, 3
- [48] W. Luo, Z. Yu, R. Rzayev, M. Satkowski, S. Gumhold, M. McGinity, and R. Dachzelt. Pearl: Physical environment based augmented reality lenses for in-situ human movement analysis. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pp. 1–15, 2023. 1, 2
- [49] L. Maister, M. Slater, M. V. Sanchez-Vives, and M. Tsakiris. Changing bodies changes minds: owning another body affects social cognition.

Trends in cognitive sciences, 19(1):6–12, 2015. 3

- [50] C. Montemayor, J. Halpern, and A. Fairweather. In principle obstacles for empathic ai: why we can't replace human empathy in healthcare. *AI & society*, 37(4):1353–1359, 2022. 9
- [51] G. A. Montes and B. Goertzel. Distributed, decentralized, and democratized artificial intelligence. *Technological Forecasting and Social Change*, 141:354–358, 2019. 8
- [52] K. Nakagaki, A. Dementyev, S. Follmer, J. A. Paradiso, and H. Ishii. Chainform: a linear integrated modular hardware system for shape changing interfaces. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*, pp. 87–96, 2016. 9
- [53] R. Newbury, K. A. Satriadi, J. Bolton, J. Liu, M. Cordeil, A. Prouzeau, and B. Jenny. Embodied gesture interaction for immersive maps. *Cartography and Geographic Information Science*, 48(5):417–431, 2021. 2
- [54] M. Nikšić, B. Tominc, and N. Goršič. Revealing residents' shared values through crowdsourced photography: Experimental approach in participatory urban regeneration. *Urbani izziv*, 29:29–42, 2018. 8
- [55] J. Nishida, S. Matsuda, H. Matsui, S.-Y. Teng, Z. Liu, K. Suzuki, and P. Lopes. Handmorph: a passive exoskeleton that miniaturizes grasp. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*, pp. 565–578, 2020. 2, 3, 9
- [56] J. Nishida, S. Matsuda, M. Oki, H. Takatori, K. Sato, and K. Suzuki. Egocentric smaller-person experience through a change in visual perspective. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1–12, 2019. 2, 3
- [57] J. Nishida and K. Suzuki. Biosync: A paired wearable device for blending kinesthetic experience. In *Proceedings of the 2017 CHI conference on human factors in computing systems*, pp. 3316–3327, 2017. 9
- [58] J. Nishida and K. Suzuki. Hyperspective: Shaping experiences beyond perspectives. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 1546–1549. IEEE, 2019. 3
- [59] A. Oulasvirta, E. Kurvinen, and T. Kankainen. Understanding contexts by being there: case studies in bodystorming. *Personal and ubiquitous computing*, 7:125–134, 2003. 9
- [60] T. C. Peck, S. Seinfeld, S. M. Aglioti, and M. Slater. Putting yourself in the skin of a black avatar reduces implicit racial bias. *Consciousness and cognition*, 22(3):779–787, 2013. 3
- [61] T. A. Pedersen, J. A. Glomsrud, E.-L. Ruud, A. Simonsen, J. Sandrib, and B.-O. H. Eriksen. Towards simulation-based verification of autonomous navigation systems. *Safety Science*, 129:104799, 2020. 9
- [62] P. Reipschläger, F. Brudy, R. Dachelt, J. Matejka, G. Fitzmaurice, and F. Anderson. Avatar: An immersive analysis environment for human motion data combining interactive 3d avatars and trajectories. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pp. 1–15, 2022. 1, 2
- [63] O. Renault, N. M. Thalmann, and D. Thalmann. A vision-based approach to behavioural animation. *The journal of Visualization and computer animation*, 1(1):18–21, 1990. 8
- [64] E. Rivers, C. Jaynes, A. Kimball, and E. Morrow. Using case study data to validate 3d agent-based pedestrian simulation tool for building egress modeling. *Transportation Research Procedia*, 2:123–131, 2014. 1, 2
- [65] J. O. Rixen, M. Colley, A. Askari, J. Gugenheimer, and E. Rukzio. Consent in the age of ar: Investigating the comfort with displaying personal information in augmented reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pp. 1–14, 2022. 6
- [66] J. O. Rixen, T. Hirzle, M. Colley, Y. Etzel, E. Rukzio, and J. Gugenheimer. Exploring augmented visual alterations in interpersonal communication. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1–11, 2021. 6
- [67] C. Sagnier, E. Loup-Escande, D. Lourdeaux, I. Thouvenin, and G. Valéry. User acceptance of virtual reality: an extended technology acceptance model. *International Journal of Human-Computer Interaction*, 36(11):993–1007, 2020. 6
- [68] E. B.-N. Sanders and P. J. Stappers. *Convivial toolbox: Generative research for the front end of design*. BIS, 2012. 2
- [69] D. Shin. Empathy and embodied experience in virtual environment: To what extent can virtual reality stimulate empathy and embodied experience? *Computers in human behavior*, 78:64–73, 2018. 3, 9
- [70] Z. Stark, L. Franzen, and A. P. Johnson. Insights from a dyslexia simulation font: Can we simulate reading struggles of individuals with dyslexia? *Dyslexia*, 28(2):228–243, 2022. 9
- [71] R. Stoakley, M. J. Conway, and R. Pausch. Virtual reality on a wim: interactive worlds in miniature. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pp. 265–272, 1995. 3
- [72] A. Tajadura-Jiménez, D. Banakou, N. Bianchi-Berthouze, and M. Slater. Embodiment in a child-like talking virtual body influences object size perception, self-identification, and subsequent real speaking. *Scientific reports*, 7(1):9637, 2017. 3
- [73] D. Thalmann. Crowd simulation. In *Encyclopedia of Computer Graphics and Games*, pp. 1–8. Springer, 2016. 9
- [74] U. Von Zadow and R. Dachelt. Giant: Visualizing group interaction at large wall displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 2639–2647, 2017. 2
- [75] J. A. Wagner Filho, W. Stuerzlinger, and L. Nedel. Evaluating an immersive space-time cube geovisualization for intuitive trajectory data exploration. *IEEE transactions on visualization and computer graphics*, 26(1):514–524, 2019. 2
- [76] N. Yee and J. Bailenson. The proteus effect: The effect of transformed self-representation on behavior. *Human communication research*, 33(3):271–290, 2007. 3
- [77] Q. Zhang, G. Barbareschi, Y. Huang, J. Li, Y. S. Pai, J. Ward, and K. Kunze. Seeing our blind spots: Smart glasses-based simulation to increase design students' awareness of visual impairment. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–14, 2022. 3